

Experiment:- Study of LAN Topology network devices and address configuration. ①

Objective:-

- 1 > To study the LAN topology used in college campus.
- 2 > To identify networking device & understand their function.
- 3 > To analyze wired & wireless communication method used in the network.
- 4 > To measure speed of network for wired & wireless connection.
- 5 > To obtain mac, IP Address & subnet mask.

• problem Statement - 1

- Draw & identify LAN topology of college campus & clearly label all components such as router, switch & etc.

Sloⁿ => the college campus network observed this study consists of:

- * Layer-3 core switches in the server room.
- * managed by POE+ Switches.
- * wireless Access point (APs)
- * Ethernet port classroom & laboratories.

Problem statement-1

→ Network Topology.

Hybrid Hierarchical topology which combines.

- i) Tree Topology.
- ii) Extended star topology.
- iii) Ring Topology. (server room).

→ The network hierarchical layers.

- ① Core layer.
- ② Distribution layer.
- ③ Access layer.

→ Core layer.

* Core layer contains server room and server room have firewall Router, 2-L3 switches, Patch, Rack, server inverter, etc.

→ Distribution layer.

* Distribution layer contains 4 switches.

* the 4 switches are

- ① main Building → R7 (Room no. 213, 2nd floor).
- ② main Build → R1 (Room no-301, 3rd floor).
- ③ main Build → R6 (Room no-113, 1st floor).
- ④ main Build → R2 (Room no-201, 2nd floor).

③
* this 4-switches contains →

① 18-port Gigabit Easy Smart switch
with 16-port [TL-SG1218MPE]

② [SN3428XMP] → 24 port Gigabit & 4-port
10GbE SFP+, L2+ managed switch with 24
port PoE.

→ Now let's discuss these 2 above
components →

① TP-Link TL-SG1218MPE

* Likely used for: CCTV camera, with
access point etc.

* 18 port Gigabit "Easy Smart" switch

* 16 port support PoE

② TP-Link Omada SN3428XMP.

* Higher-end managed layer 2 switch.

* 24 PoE port, uplinks, 10GbE

* usually acts as a main distribution
for each floor.

Access layers: -

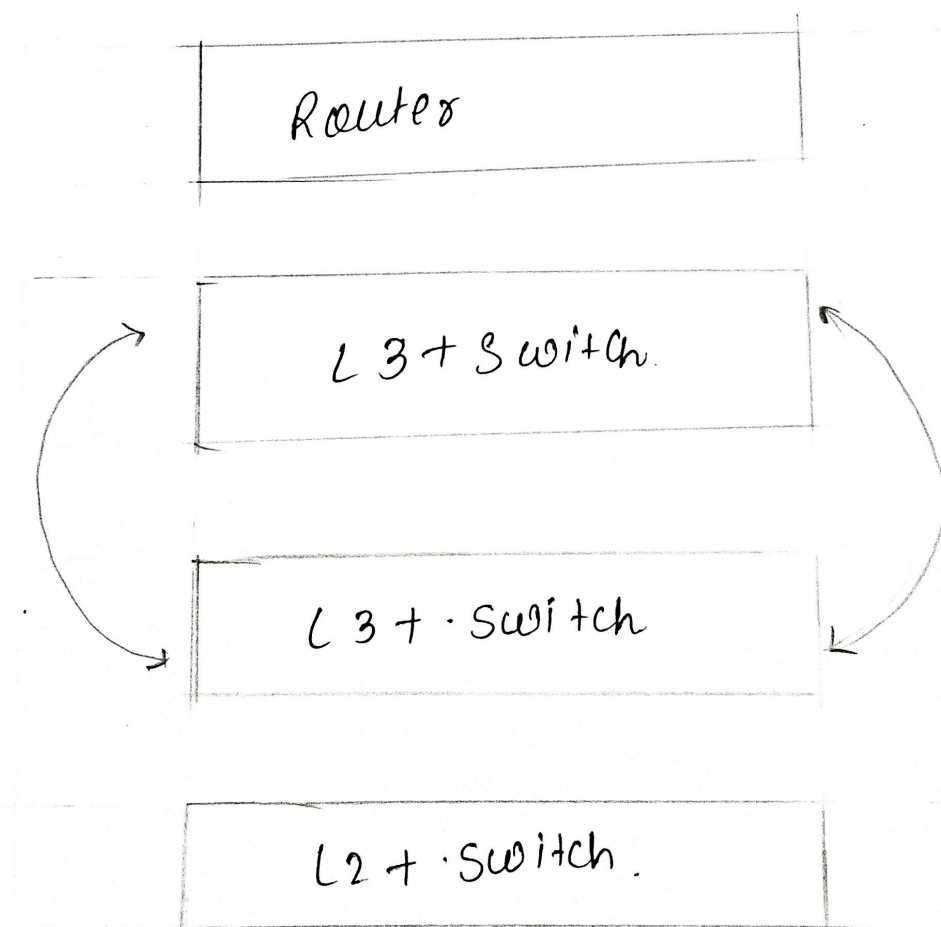
- Access layer R1 [Room no- 301, 3rd floor] (main Build R1).
- main Build R4 [Room no- 05, ground floor].
- main Build R5 (Room no- 18, ground floor).

* Every switch acts as an access layer switch which helps to provide network connectivity to all classroom, lab, faculty room, etc.

* Access layer directly connects:

- ① Ethernet port
- ② wireless AP
- ③ lab system
- ④ outdoor wireless AP (ex-08)

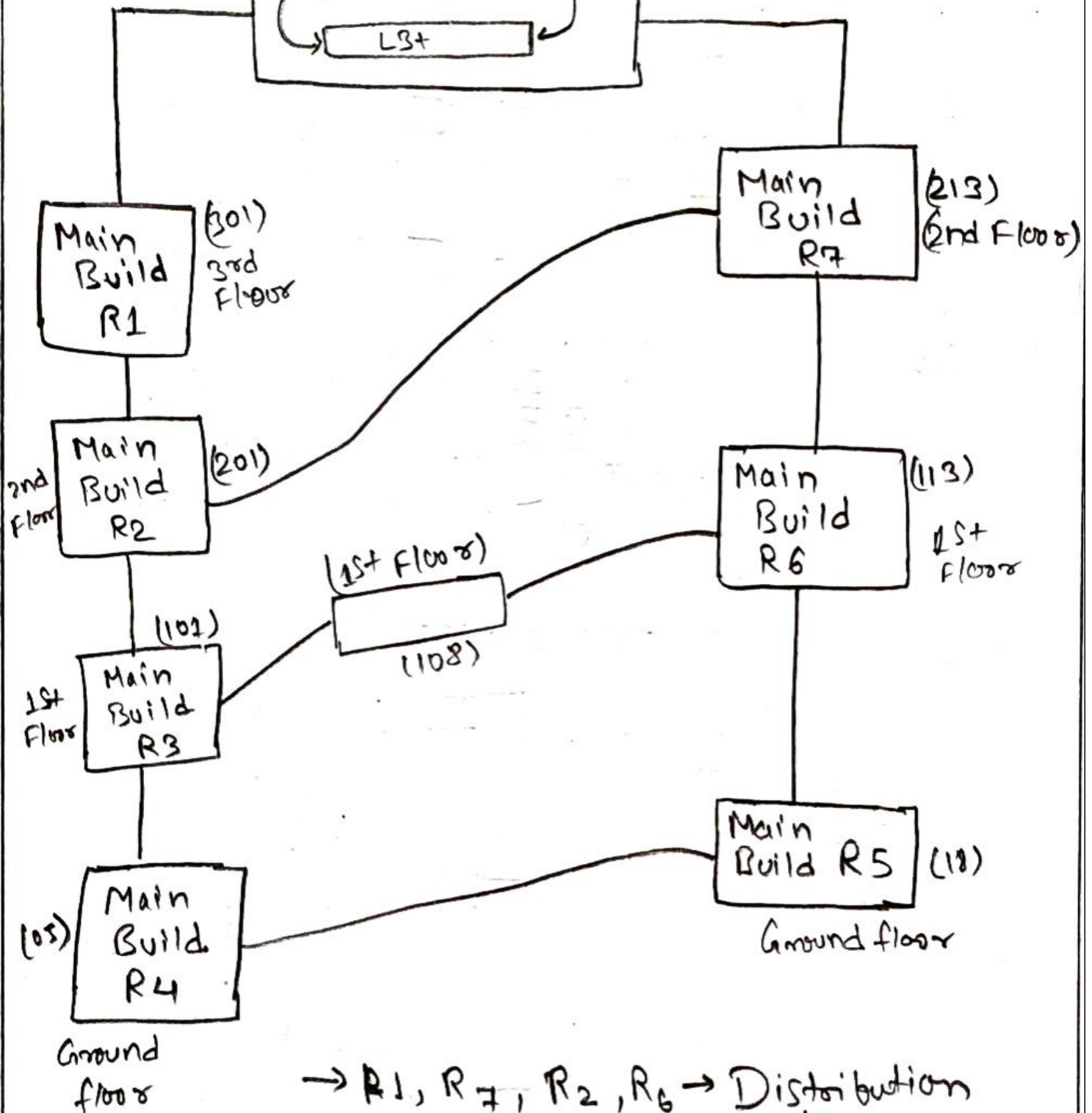
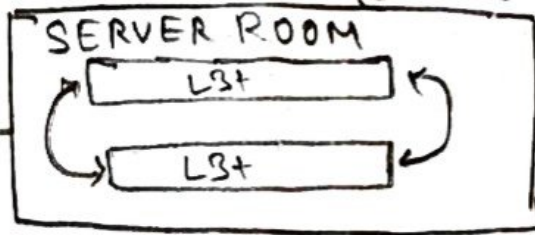
→ server room Architecture/topology.



TOPOLOGY

⑤

(210) (2nd Floor)



→ R1, R7, R2, R6 → Distribution layer

→ Server room → Core layer

→ R1, R2, R3, R4, R5, R6, R7 → Access layer

```
C:\Users\Hp>netsh wlan show interfaces
```

```
There is 1 interface on the system:
```

```
Name                : Wi-Fi
Description          : Intel(R) Wi-Fi 6E AX211 160MHz
GUID                 : b8024ae2-5fa3-414e-84b0-27e3ab35aeb6
Physical address     : dc:45:46:4a:5a:1a
Interface type       : Primary
State                : connected
SSID                 : Redmi 15 5G
AP BSSID             : 3e:73:f1:e1:b3:bf
Band                 : 2.4 GHz
Channel              : 1
Connected Akm-cipher : [ akm = 00-0f-ac:02, cipher = 00-0f-ac:04 ]
Network type         : Infrastructure
Radio type           : 802.11n
Authentication       : WPA2-Personal
Cipher               : CCMP
Connection mode      : Auto Connect
Receive rate (Mbps)  : 72.2
Transmit rate (Mbps) : 72.2
Signal               : 96%
Rssi                 : -27
Profile              : Redmi 15 5G
QoS MSCS Configured  : 0
QoS Map Configured   : 0
QoS Map Allowed by Policy : 0
```

```
C:\Users\Hp>
```

Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.6.87.2-microsoft-standard-WSL2 x86_64)

* Documentation: <https://help.ubuntu.com>
* Management: <https://landscape.canonical.com>
* Support: <https://ubuntu.com/pro>

System information as of Thu May 28 14:12:09 UTC 2026

System load:	0.8	Processes:	56
Usage of /:	0.2% of 1006.85GB	Users logged in:	0
Memory usage:	10%	IPv4 address for eth0:	172.25.86.245
Swap usage:	0%		

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s just raised the bar for easy, resilient and secure K8s cluster deployment.

<https://ubuntu.com/engage/secure-kubernetes-at-the-edge>

This message is shown once a day. To disable it please create the `/home/komal/.hushlogin` file.

```
komal@LAPTOP-EDN1VAID:~$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
    inet 172.25.86.245  netmask 255.255.240.0  broadcast 172.25.95.255
    inet6 fe80::215:5dff:fe88:278c  prefixlen 64  scopeid 0x20<link>
    ether 00:15:5d:88:27:8c  txqueuelen 1000  (Ethernet)
    RX packets 310  bytes 375544 (375.5 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 98  bytes 9058 (9.0 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0
```

```
lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
    inet 127.0.0.1  netmask 255.0.0.0
    inet6 ::1  prefixlen 128  scopeid 0x10<host>
    loop txqueuelen 1000  (Local Loopback)
    RX packets 20  bytes 2436 (2.4 KB)
    RX errors 0  dropped 0  overruns 0  frame 0
    TX packets 20  bytes 2436 (2.4 KB)
    TX errors 0  dropped 0  overruns 0  carrier 0  collisions 0
```

```
komal@LAPTOP-EDN1VAID:~$ ifconfig /all
```

Performance

Run new task

**CPU**
2% 1.88 GHz

**Memory**
6.6/7.6 GB (87%)

**Disk 0 (C: D:)**
SSD (NVMe)
4%

**Ethernet**
Ethernet 3
S: 0 R: 0 Kbps

**Wi-Fi**
Wi-Fi
S: 0 R: 0 Kbps

**GPU 0**
Intel(R) UHD Graphics
1%

**GPU 1**
NVIDIA GeForce RTX...
0% (43 °C)

Wi-Fi

Intel(R) Wi-Fi 6E AX211 160MHz



 Send

 Receive

0 Kbps

0 Kbps

Adapter name: Wi-Fi

SSID: Redmi 15 5G

Connection type: 802.11n

IPv4 address: 10.175.187.37

IPv6 address: 2401:4900:73df:b789:882a:3731:aff5:2f7e

Signal strength: 



Performance

Run new task

CPU
39% 2.60 GHz

Memory
6.7/7.6 GB (88%)

Disk 0 (C: D:)
SSD (NVMe)
67%

Ethernet
Ethernet 3
S: 0 R: 0 Kbps

Ethernet
Ethernet 4
S: 368 R: 8.0 Kbps

GPU 0
Intel(R) UHD Graphics
9%

GPU 1
NVIDIA GeForce RTX...
0% (43 °C)

Ethernet

Throughput

Remote NDIS Compatible Device

54 Mbps



60 seconds

Send
368 Kbps
Receive
8.0 Kbps

Adapter name: Ethernet 4
Connection type: Ethernet
IPv4 address: 10.156.162.70
IPv6 address: fe80::ed76:cced:3d8f:ba0c%62



```
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```
There is 1 interface on the system:
```

Name	: Wi-Fi
Description	: Intel(R) Wi-Fi 6E AX211 160MHz
GUID	: b8024ae2-5fa3-414e-84b0-27e3ab35aeb6
Physical address	: dc:45:46:4a:5a:1a
Interface type	: Primary
State	: disconnected
Radio status	: Hardware On Software On

(10)
- Difference b/w wired and wireless speed?

Soln
wired network

- * faster communication
- * stable connection
- * low latency
- * Dedicated physical medium

wireless network.

- * slower communication
- * Affected by distance / Interference.
- * Higher latency
- * shared radio medium.

→ Ethernet provides direct physical communication using network cables resulting in higher speed & stable performance.

→ wireless connections use radio waves which are affected by interference, distance, obstacles & multiple connected users.

① Role of subnet mask in network?

→ Soln

A subnet mask identifies the network portion & host portion of an IP address.

It helps devices determine whether another device belongs to the same network or different network.

* It is important for:

- ① Network segment
- ② Efficient routing
- ③ Communication b/w devices.

— OBSERVATION.

During the study of the college network infrastructure :-

- ① A centralized server room was used as network backbone.
- ② Two L3 switches were connected in redundant cyclic structure.
- ③ Managed TP-link PoE+ switching were used in network cabinets.

→ Advantage of Existing network?

- * scalable architecture
- * centralized management
- * supports PoE device.
- * Easy expansion of floors & classroom.
- * support both wireless & wired
- * Redundant link for failover.

Conclusion:—

The observed campus LAN follow a hybrid hierarchical topology consist of Tree Extended Star Ring. In this experiment, we studied LAN topology identified different network devices and learned how to find IP Address, MAC address and subnet mask. we also compared wired and wireless connection and understood their performance differences.